CSE 4345 – Big Data Analytics
Complex Engineering Problem (CEP) Assignment

# EpiScope

A Big Data Surveillance, Forecasting and Response-Planning Platform
for Seasonal, Tropical and Epidemic Diseases

**Group [GROUP NUMBER]**


| Member | Student ID | Lead role |
|---|---|---|
| [MEMBER 1 NAME] | [ID] | Data Engineer |
| [MEMBER 2 NAME] | [ID] | Analytics Engineer |
| [MEMBER 3 NAME] | [ID] | Visualisation Engineer |


Repository: [https://github.com/<owner>/<repo> (private, shared with the instructor, tag v1.0)]
Video: [shareable video link]


Department of Computer Science & Engineering, Premier University, Chittagong – Spring 2026
Instructor: Rokan Uddin Faruqui


All encounter, facility and stock data in this report are **synthetic** (teaching case); no real patient data were used.

# EpiScope: A Big Data Surveillance, Forecasting and Response-Planning Platform for Bangladesh

Group [N]: [Member 1], [Member 2], [Member 3]
Dept. of CSE, Premier University, Chittagong, Bangladesh

***Abstract*—EpiScope is an end-to-end prototype that turns routine health-facility encounters, climate, population and logistics data into district-level early warnings, 1–4-week forecasts of cases and hospital demand, an optimal allocation of a fixed national stock, and a seasonal preparedness calendar, served through a privacy-preserving D3.js dashboard. A seeded generator produces 28,239,687 synthetic encounters (SF = 1, 2021–2025, 64 districts, 786 facilities, 35 injected outbreaks and reporting artefacts) driven by weather; Apache Spark builds a Delta Lake medallion lakehouse with 27 quality rules, quarantine, salted hashing, idempotent incremental loads and row-level lineage. A seasonal CUSUM detector finds 91% of injected local outbreaks at 2.02 false alerts per district-year; per-district negative-binomial GLMs reduce the weighted interval score to 0.36 of a seasonal-naive baseline one week ahead; a mixed-integer programme with an explicit equity floor leaves less weighted unmet demand than a population-proportional plan. The full pipeline rebuilds from a clean clone with one command in 22.5 minutes on a 4-core, 16 GB machine.**

***Index Terms*—disease surveillance, Apache Spark, Delta Lake, outbreak detection, forecasting, optimisation, privacy, D3.js**

## I. Introduction and Problem Analysis

Bangladesh carries a heavy and strongly seasonal infectious-disease burden: dengue follows the monsoon (2023 was the worst season on record), acute watery diarrhoea and cholera rise before the monsoon and after floods, influenza-like illness (ILI) peaks in the rainy season, and malaria is concentrated in the Chittagong Hill Tracts. The data needed to act early exist but are scattered across facility reporting (DHIS2-style), laboratories, the meteorological department, the census and district stores, with different formats, delays and quality. The client brief asks for a platform that, for every district, (i) detects unusual rises early, (ii) forecasts cases and hospital demand 1–4 weeks ahead, (iii) recommends how to allocate a limited national stock and rapid-response teams, and (iv) produces a preparedness calendar – explainably, robustly to late and incomplete data, on modest infrastructure, and without exposing identifiable patients.

**Requirements.** We translated the brief into the catalogue of 35 requirement IDs (ING, DQ, PRV, AN, VIS, ENG) and tracked every one in the repository README (ID → file → report section). All 21 Must items and all 11 Should items are implemented; of the 5 Could items we implemented ING-7 (Delta time travel), PRV-4 (k-anonymity), AN-10 (SEIR) and VIS-7 (mobile layout and animated playback).

**Stakeholders** (Table I) have conflicting needs, which drive five explicit conflicts (C1–C5, Section II). The key analytical difficulty is *interdependence*: ingestion quality determines detection, detection and forecasts feed the optimiser, and the optimiser's uncertainty must be visible in the dashboard, so an error in one stage propagates through all of them.

TABLE I
Stakeholders, needs and how EpiScope serves them

| Stakeholder | Primary need / concern | EpiScope feature |
|---|---|---|
| MoH policymakers | national picture; defensible allocation ("why X over Y?") | optimiser with stated weights, baseline comparison, policy brief |
| Epidemiologists | reliable, transparent signals | two methods, trade-off curve, artefact guard, evaluation panel |
| District officers | early local warning; alert fatigue | ranked table, per-district forecasts, district role view |
| Hospital managers | bed/consumable demand with uncertainty | demand quantiles, 50/80/90% bands |
| Vector control | timing of larval campaigns | preparedness calendar (onset − 8 weeks) |
| Patients & public | no misuse of data | salted hashing, suppression, aggregates only |
| Privacy regulator | minimisation, small-cell suppression | PRV-1..4, audit log, role tokens |

## II. CEP Attributes and Conflict Resolution

Table II maps each Complex Engineering Problem attribute to concrete design decisions.

**C1 sensitivity vs. specificity.** We evaluated four detectors over a threshold grid against the ground-truth file (which the detector never reads) and chose the operating point CUSUM $\geq 17.0$ with the rule "highest sensitivity with at most three false alert events per district-year over the four diseases" – fewer than one per disease per year, a workload a district officer can verify by phone (Section VI-A). **C2 granularity vs. privacy.** Upazila maps would be more actionable but 1–4-case cells identify people; we publish district×epi-week counts with primary and complementary suppression, and only aggregates for districts a user is not responsible for. **C3 efficiency vs. equity.** The equity floor $\alpha$ guarantees each district at least $\alpha$ times its population share of every resource shipped; the curve in Section VI-C shows what each $\alpha$ costs. **C4 freshness vs. cost.** Allocation and forecasting decisions are weekly and reporting delay is days, so a daily incremental batch is the main cadence; a Structured Streaming path refreshes alerts within five minutes for emergencies at the cost of a long-running process. **C5 accuracy vs. interpretability.** The pooled

TABLE II
HOW THE SOLUTION DEALS WITH EACH CEP ATTRIBUTE (P1–P7)

| | |
|---|---|
| P1 Depth of knowledge | Distributed processing (Spark SQL, window functions, grouped pandas UDFs, AQE, broadcast joins), columnar ACID storage (Delta/Parquet, partition pruning, column pruning), surveillance statistics (EARS C1–C3, historical limits, CUSUM), probabilistic forecasting (GLM with negative-binomial dispersion, WIS), mixed-integer optimisation (HiGHS), STL decomposition, SEIR modelling, privacy (salted hashing, complementary suppression, k-anonymity), visual encoding and interaction (focus+context, linked brushing, colour-blind-safe palettes). |
| P2 Conflicting requirements | Five conflicts resolved with numbers, not opinion: C1 trade-off curve and operating point (§VI-A); C2 district×week aggregation with primary+complementary suppression (§IV-F); C3 efficiency–equity curve with a recommendation rule (§VI-C); C4 daily batch + 5-minute streaming path (§III-C); C5 GLM vs gradient boosting backtest (§VI-B). |
| P3 Depth of analysis | No single correct answer: thresholds, horizons, lag windows (rain 4–8 weeks), dispersion estimation, the equity floor $\alpha$ and the planning quantile were chosen by backtests and sensitivity analysis against baselines (EARS vs CUSUM, GLM vs seasonal naive vs HGB, optimiser vs population share). |
| P4 Unfamiliar issues | Reporting delays (0–21 days) handled by a reporting triangle and nowcast; late records update past epi-weeks (MERGE + windowed recompute); a facility that goes silent for 3 weeks and back-fills, and a large facility that starts reporting mid-stream, are both injected and handled; 2% resubmitted records and cross-facility duplicate visits are de-duplicated; Dhaka-driven skew handled by broadcast joins and AQE. |
| P5 Beyond standard codes | No standard maps these data to an allocation plan: we combined WHO IDSR alert/action logic, ICD-10 coding, data-protection principles (minimisation, purpose limitation, suppression) with our own formulation (valued unmet demand, equity floor, integer teams). |
| P6 Stakeholders | Seven stakeholder groups with different needs (Table I); role-based views (national vs district), plain-language policy brief (Appendix A), usability protocol with non-CS users (§VII). |
| P7 Interdependence | One code path end to end: the same silver rules feed detection, forecasting and streaming; detection sets optimiser weights and team demand; forecast quantiles set planning demand; every gold row carries the bronze batch IDs it came from, so a bad input batch can be traced to every output it touched. |

gradient-boosting model is slightly more accurate one to three weeks ahead (relative WIS 0.33 vs 0.36 at one week) but not at four weeks, and over-covers its intervals; the per-district GLM, whose coefficients (lagged rain, temperature, last season) officials can read and challenge, is within about 10% and is the operational model, with the boosted model kept as a monitored challenger.

## III. Architecture and Design Decisions

Fig. 1 shows the medallion lakehouse, analytics and serving layers; Table III lists the main alternatives we considered.

### A. Processing and storage

All heavy work runs in PySpark 3.5 (`ING-2`); pandas appears only inside grouped UDFs on per-district series and for final aggregated exports. Every layer is a **Delta Lake** table (Parquet + transaction log) registered in a Hive catalog backed by Derby (`ING-3`): `episcope_bronze`, `_silver` and `_gold` databases. Delta gives ACID MERGE (idempotent upserts), `replaceWhere` (recompute only affected epi-weeks) and time travel (`ING-7`). Setting `paths.lake_root` to an `s3a://` URI switches storage to MinIO through the S3A connector (`ING-5`; MinIO service in `docker-compose.yml`).

### B. Partitioning

Silver encounters are partitioned by `(epi_year, adm1_pcode)`: almost every query restricts the period, and allocation and division dashboards restrict the division; 6 years × 8 divisions gives 48 directories whose files stay large (tens of MB) at SF = 1. Partitioning by district would create 384 directories with many small files and skew (Dhaka alone holds a large share of the rows); by epi-week it would create thousands of tiny files. Gold marts are small and partitioned by `epi_year`; bronze is partitioned by load batch so that silver reads exactly one batch.

### C. Cadence (C4) and streaming

The operational cadence is a daily incremental batch (`make drop`); decisions it supports (weekly forecasts, fortnightly shipments) do not need fresher data, and the batch is cheap. For emergencies a Structured Streaming job (`ING-6`, `pipeline/streaming/stream_alerts.py`) watches a file inbox and, in each micro-batch, reuses the *same* bronze/silver/gold code and detection, refreshing `alerts.json`; the file-to-alert latency is logged (§VI-E).

### D. Reproducibility and orchestration

`make all` creates a virtual environment from a fully pinned `requirements.txt`, writes a git-ignored `.env` with a random salt and role tokens, and runs a small DAG runner (`orchestration/dag.py`): tasks with dependencies, one fresh JVM per stage, retries with exponential back-off and a JSON run record with per-task timings (`ENG-1`, `ENG-4`). A Dockerfile and `docker-compose.yml` give the same result in containers. Nothing hard-codes dates, file names or row counts: files are recognised by their header and the as-of date is the latest report date in silver.

## IV. Data Generation, Quality, Lineage and Privacy

### A. Public data (S1–S3, S7)

District (ADM2, 64) and upazila (ADM3, 507) boundaries and centroids come from OCHA COD-AB v03 (HDX), simplified to a 55 kB TopoJSON for the browser. Weather is requested from the Open-Meteo archive at each district centroid (rate-limited, responses cached unmodified in the landing zone and ingested as raw bytes), population by age and sex from the HDX COD-PS table (BBS census 2022), and the OpenDengue V1.3 Bangladesh series is used to check the synthetic seasonality. Each source records its status in a

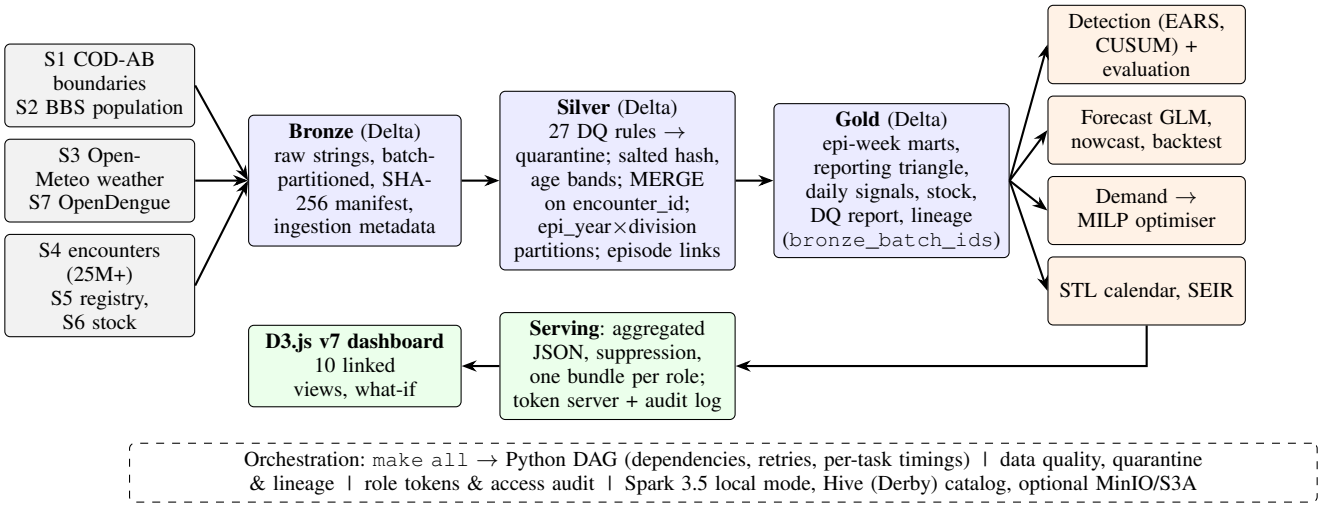



Fig. 1. EpiScope architecture (medallion lakehouse + analytics + D3.js serving layer).

TABLE III
ALTERNATIVES CONSIDERED

| Decision | Chosen | Alternative (why not) |
|---|---|---|
| Table format | Delta Lake | plain Parquet (no MERGE/time travel); Iceberg (heavier catalog set-up) |
| Storage | local FS / MinIO | HDFS in Docker (more RAM on a laptop) |
| Orchestration | Makefile + DAG runner | Airflow/Prefect (services and memory for a 6-task DAG) |
| Detection | CUSUM on historical limits | Farrington flexible (needs years of clean baseline per series; slower) |
| Forecast | per-district NB GLM | pooled HGB (less accurate here, opaque); ARIMA (no count noise, weather lags awkward) |
| Optimiser | SciPy HiGHS MILP | PuLP/CBC (extra binary); heuristics (no optimality guarantee) |
| Serving | stdlib server + static JSON | database-backed API (not needed for $<$10 MB of aggregates) |

provenance file shown on the dashboard: *real*, *reference* (committed real snapshot) or *synthetic fallback* when a provider is unreachable. In the build reported here boundaries were real, weather synthetic fallback and population synthetic fallback (the build machine's network policy blocked Open-Meteo and HDX); re-running `make all` with internet access replaces the fallbacks, and every number in this report is regenerated by `make report`.

### B. Synthetic encounter generator (S4–S6)

`generator/generate.py` is parameterised by a scale factor and a seed and is deterministic: each (district, year) chunk has its own `SeedSequence`, so output is byte-identical for any number of worker processes (tested). At SF = 1 it writes 28,239,687 encounter rows reported in 2021–2025 plus 194,626 rows in 14 future daily drops, a registry of 786 facilities placed on real upazilas (district hospitals, upazila health complexes, medical colleges, Dhaka-heavy private clinics) and daily JSON stock and bed-occupancy snapshots. Expected daily counts per district and disease are: dengue driven by 4–8-week lagged rainfall and 3–6-week temperature suitability, amplified in large cities; diarrhoea with pre- and post-monsoon peaks and spikes 3–14 days after extreme rainfall; ILI with a rainy-season and a winter peak; malaria concentrated in the hill districts. Counts are drawn from a negative binomial (Poisson–gamma). We injected 35 outbreaks: 22 sharp local outbreaks of different size and duration, 3 slow-growing ones, a 2023-like dengue epidemic in five cities, post-flood cholera in Sylhet, two outbreaks that begin in the future drops, and a nationwide novel-respiratory surge spreading from Dhaka with distance. Realism defects: ≈2% resubmitted `encounter_id`s, 0.5% orphan facility keys, ≈1% invalid values, a right-skewed 0–21-day reporting delay with fewer reports on Fridays, Saturdays and holidays, a facility silent for 21 days that then back-fills, and a large private network that starts reporting mid-stream. Truth files (outbreaks, artefacts, duplicate pairs) are written to `data/ground_truth`, which only the evaluation module reads (enforced by a test).

### C. Ingestion and incremental, idempotent loads

Bronze (`ING-1`) stores every row unmodified as strings plus `_source, _load_ts, _batch_id, _file_name, _file_checksum`; CSV columns are matched by name, so re-ordered or extra columns are tolerated. The SHA-256 of every file is kept in a manifest, and a file already seen is skipped. Silver MERGEs on `encounter_id` (keeping the earliest report date as `first_report_date`); gold is recomputed with `replaceWhere` from the Monday of the earliest onset week in the batch. Hence re-running a day changes nothing and late records (up to 21 days, and back-filled ones beyond) update the epi-weeks they belong to (`ING-4`); an integration test loads a drop, reloads it, force-reloads it and checks that silver and gold are identical.

### D. Data quality (DQ-1, DQ-2, DQ-4)

27 rules are evaluated as Spark column predicates (16 on encounters: schema, IDs, referential integrity, dates, onset≤report, plausible delay, date range, age, sex, ICD-10 format and category, booleans, outcome, geography; 6 on the registry; 5 on stock). A row failing any rule goes to `silver.quarantine` with all its reason codes; the per-load report (rows in/out, quarantined, duplicates, pass rate per rule) is a gold table and a dashboard view. In the SF = 1 load 440,423 of 28,239,687 encounter rows (1.56%) were quarantined and 544,599 resubmissions removed. Cross-facility duplicate visits (`DQ-4`) are linked when the salted hash, disease group, sex and adjacent age band agree and onsets are at most 7 days apart; against 127,264 injected duplicate pairs (3% with a wrong sex, 30% with a mistyped age) precision was 100.00% and recall 97.1%, and 0 genuinely separate episodes were merged.

### E. Lineage (DQ-3)

Every gold row carries `bronze_batch_ids`, the sorted set of bronze batches whose records produced it, and `gold.lineage_batches` maps each batch to its files, checksums and row counts; a test checks that every contributing batch exists in the manifest.

### F. Privacy by design (PRV-1..4)

Patient IDs are replaced in silver by SHA-256(salt ‖ ID) with a salt that lives only in the git-ignored `.env`; exact ages become six bands; the quarantine stores hashes too (`PRV-1`). The browser receives only aggregates: primary suppression hides every count from 1 to 4, and complementary suppression hides a second cell whenever a published district→division→national total would reveal a single hidden cell (iterated to a fixed point); no totals across weeks or diseases are published. An automated audit replays every exported hierarchy and scans every count field (`PRV-2`). Role tokens select a bundle: national users see all districts, a district user sees their district in detail and everything else only as division aggregates (`PRV-3`); each access is logged. A k-anonymity check on a record-level research extract (`PRV-4`, Table IV) climbs a generalisation ladder until $k \geq 5$ with $\leq 1\%$ suppression. For public releases we recommend differential privacy instead: adding Laplace noise with scale $1/\varepsilon$ to weekly district counts (sensitivity 1 per person-week) protects against linkage attacks that k-anonymity cannot stop, at the cost of bias in small districts; $\varepsilon \approx 1$ per released table and a privacy budget per year are reasonable starting points.

TABLE IV
K-ANONYMITY OF A MALARIA RESEARCH EXTRACT (K = 5)

| Level | Records | Classes | $k_{\min}$ | Suppressed | ≤1% |
|---|---|---|---|---|---|
| L0 | 92,249 | 18868 | 1 | 24.7% | no |
| L1 | 92,249 | 8390 | 1 | 10.0% | no |
| L2 | 92,249 | 3342 | 1 | 4.1% | no |
| L3 | 92,249 | 2066 | 1 | 2.5% | no |
| L4 | 92,249 | 871 | 1 | 0.8% | yes |
| L4-released | 91,532 | – | 5 | 0.8% | yes |

L0 district×week×6 age bands×sex; L1 month; L2 division; L3 3 age bands; L4 quarter.

## V. Analytics Methods

### A. Outbreak detection (AN-1)

Detection runs daily per district and disease on counts by first report date. The *guarded* signal excludes duplicate episodes and back-filled records (records reported on the first day a normally daily facility reports again after ≥7 silent days, with delay ≥3 days); days on which more than 15% of the records come from a facility that started reporting in the last 56 days are flagged "verify reporting" rather than alerted. Weekday reporting effects are removed with national weekday factors. **EARS C1/C2/C3** [1] use Spark window functions: $z_t = (x_t - \bar{x})/s$ over a 7-day baseline (C2/C3 with a 2-day guard band; C3 sums positive C2 exceedances over 3 days), with $s \geq \max(\sqrt{\bar{x}}, 0.5)$ for sparse series. The **seasonal CUSUM** runs in a grouped pandas UDF: the expected 7-day count is the historical limit [3] (mean of the same window ±2 weeks in each earlier year) multiplied by this year's level over the last 2–7 weeks (2-week guard band), the daily count is standardised by $\sqrt{\mu\hat{\phi}}$ with a prospective quasi-Poisson dispersion $\hat{\phi}$ (mean squared Pearson residual over the past year, winsorised), and $C_t = \max(0, C_{t-1} + z_t - 0.5)$ [4]. An alert also needs at least 5 cases in the last 7 days.

### B. Forecasting and nowcasting (AN-4)

For every district and disease and each horizon $h \in \{1..4\}$ weeks, a direct log-linear Poisson model (ridge $\lambda = 0.02$) uses the last two weeks (log), the mean of weeks $t - 5..t - 2$, the same season last year, annual harmonics of the target week, rainfall summed and temperature averaged over weeks $t + h - 8..t + h - 4$ (the 4–8-week lag), and the recent maximum daily rainfall. A negative-binomial size estimated by moments gives predictive quantiles. The 256 series are trained in parallel with `groupBy(district, disease).applyInPandas`. Because the last 3–4 weeks are incomplete, the counts known at the origin are rebuilt from the reporting triangle (onset week×report week) and divided by the national completeness at that delay, estimated only from weeks 8–60 weeks before the origin. Baseline: seasonal naive (same week last year) with empirical error quantiles; comparator: gradient-boosted trees pooled over districts on $\log(1 + y)$.

### C. Demand conversion (AN-6)

Weekly case quantiles are sampled (1,000 Monte-Carlo draws by inverse-CDF interpolation), admissions are binomial with the district's admission rate over the last 52 weeks shrunk to the national rate (50 pseudo-cases), and converted with stated assumptions: 6 bags of IV saline per admitted dengue case and 8 per admitted diarrhoea case (+0.3 per outpatient diarrhoea case), 1.3 NS1 kits per suspected dengue case, 10 ORS sachets per diarrhoea case, 1 artemisinin course per

malaria case, and length of stay 3–5 days for bed-days. The mean and 10/50/80/90% quantiles over 4 weeks are kept.

### D. Allocation optimiser (AN-7)

With $\hat{y}_{d,r}$ the 80th percentile of demand, $s_{d,r}$ the latest stock, $c_r$ unit cost, $p_d$ population share and $w_d = 1 + [\text{alert}] + 0.5[\text{forecast growth} > 20\%]$:

$$\begin{aligned}
&\min \ \textstyle\sum_{d,r} w_d h_r (u_{d,r} + \beta q_{d,r}) + \sum_d w_d H_T v_d + \epsilon \sum_{d,r} c_r x_{d,r} \\
&\text{s.t. } u_{d,r} \geq \hat{y}_{d,r} - s_{d,r} - x_{d,r}, \quad q_{d,r} \geq u_{d,r} - g_{d,r}/2, \\
&\quad \textstyle\sum_{d,r} c_r x_{d,r} \leq B, \\
&\quad x_{d,r} \geq \alpha p_d \textstyle\sum_{d'} x_{d',r}, \quad \textstyle\sum_{d \in k} \sum_r m_r x_{d,r} \leq T_k, \\
&\quad v_d \geq \tau_d - z_d, \ \textstyle\sum_d z_d \leq N, \ z_d \in \mathbb{Z}_{\geq 0}, \ x, u, v \geq 0.
\end{aligned}$$

Unmet units are valued in patient-equivalents $h_r$ = share of one patient's need a unit covers $\times$ a severity weight (life-threatening 3, severe 2, moderate 1, diagnostic 0.3): IV bag $\frac{1}{6} \times 3 = 0.5$, ACT course $1 \times 2 = 2$, ORS sachet $\frac{1}{10} \times 1 = 0.1$, NS1 kit $\frac{1}{1.3} \times 0.3 = 0.23$; a missing team $H_T = 20$. These weights must be agreed with clinicians; they make sachets, kits and courses commensurable. The part of a shortfall beyond half of the district's gap $g_{d,r}$ costs $(1+\beta)$ times more ($\beta = 1$): the first half of every gap – the life-saving minimum – is filled first. This convex penalty also removes a degeneracy we observed: with purely linear valuation every taka has the same value in every district, and the solver split ties arbitrarily (a high-need district could get nothing). $\tau_d$ is the number of diseases alerting in district $d$ (rapid-response team demand, $N = 20$ teams); $m_r$ are unit weights and $T_k$ a truck-capacity limit per division. $\alpha$ is the policy lever of conflict C3: the guaranteed fraction of each district's population share of every resource shipped. The MILP ($\approx$640 variables) is solved with HiGHS [13] in well under a second. Baseline: the same budget split across resources by national gap and across districts by population.

### E. Preparedness calendar (AN-9)

Weekly counts per division and disease are decomposed with robust STL [12] (period 52, log scale) in a grouped UDF; the seasonal component averaged by ISO week gives a multiplicative seasonal index. The season is the longest circular run of $\geq$3 weeks with index $\geq 1.15$; campaigns start 8 weeks (dengue larval control), 6 (malaria) or 4 weeks before onset, supplies are pre-positioned 4 weeks before onset and staff surge 2 weeks before the peak, projected onto the next 52 weeks.

### F. Epidemic scenarios (AN-10)

The novel-respiratory surge is located from data alone (national ILI above 130% of the same-week median of other years). A deterministic SEIR model (latent 3 days, infectious 5 days) is fitted to the weekly excess by least squares on the square-root scale (free: $\beta$, $I_0$, reporting fraction $\rho$), giving $R_0 = 2.14$, $\rho = 0.25\%$ and $R^2 = 0.990$. What-if scenarios reduce $\beta$ by 20% or 40% from the detection week, two weeks earlier or later (Table X).

## VI. Evaluation

All results are from the SF = 1 build (28,239,687 encounters), evaluated prospectively over 2022-01-01–2025-12-31 (the first year is the baseline burn-in).

### A. Detection (AN-2, AN-3, C1)

Definitions: an outbreak (district$\times$disease) is detected if an alert falls between 3 days before its start and 7 days after its end; days to detection count from the start; an *alert event* is a run of consecutive alert days, false if it overlaps no outbreak window. Local outbreaks (96 rows including the 64 district rows of the national surge) and the surge are reported separately. Fig. 2 and Table V give the trade-off. At the chosen operating point CUSUM $\geq 17.0$ the pipeline detects 91% of local outbreaks with a median of 15 days after onset at 2.02 false alerts per district-year (all four diseases together), and flags 100% of districts in the national surge (median 95 days). **Reporting artefacts (AN-3).** In the back-fill district and window the naive signal produced 15 alert days against 0 for the guarded pipeline; around the onboarding facility, 0 vs 0 at the operating point (at lower thresholds the new-facility flag routes such days to "verify reporting" instead of "alert"). **Interpretation.** The seasonal CUSUM dominates the EARS family everywhere on the curve: at a comparable false-alert rate (about 3 per district-year) EARS C3 finds 59% of local outbreaks and EARS C1 12%, CUSUM over 90%. EARS compares a day with the previous week only, so every seasonal upswing and every post-flood spike looks anomalous; CUSUM compares with the same season in earlier years and accumulates small persistent excesses. The price is delay: 15 days at the operating point versus 3–5 days at thresholds that produce 10–20 false alerts per district-year – this is conflict C1 in numbers, and the what-if slider lets epidemiologists move along it. The national surge is detected in every district, but late relative to its injected start (95 days): the start is defined as the day seeding began, when only a few infections per million existed, so the first weeks carry no visible excess. Three design errors were found and fixed by this evaluation (AI_USAGE.md): a too-narrow level adjustment that saturated CUSUM in epidemic years, CUSUM increments computed on overlapping 7-day sums (strongly autocorrelated, which inflates false alerts), and a median-based dispersion estimate that under-estimated variance.

### B. Forecasting backtest (AN-5, C5)

Rolling-origin backtest over the last 52 complete weeks (models refitted every 4 origins on data complete at the origin; nowcast inputs), scored on identical cells for all models (Table VI). Averaged over all districts and diseases, the GLM's one-week-ahead MAE is 31.7 vs 62.2 for the seasonal naive and 26.4 for pooled boosting; its WIS is 0.36 of the baseline's at one week and 0.68 at four weeks, and 80% intervals cover 83% (h = 1) and 83% (h = 4) of outcomes, close to the nominal 80% (seasonal naive: 81%). Skill decays with horizon as expected; the largest gains are for ILI and dengue, whose

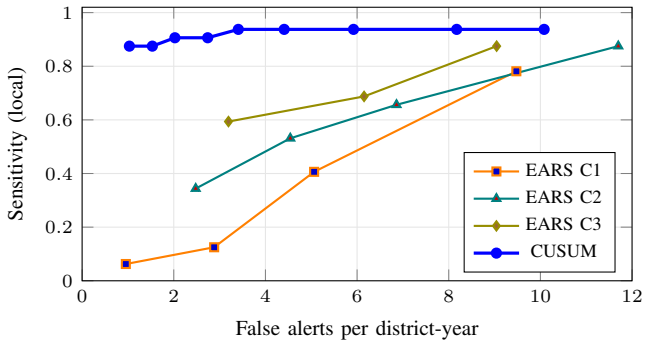



Fig. 2. Detection trade-off (guarded pipeline), one point per threshold.

weather-driven and multi-year patterns the naive method cannot follow. The pooled boosted model is slightly better one to three weeks ahead (relative WIS 0.33 at $h = 1$) but its intervals over-cover (too wide) and it loses the advantage at four weeks; given a difference of about 10% we keep the GLM, whose coefficients can be shown to officials (C5), and log the challenger for review.

### C. Allocation vs. baseline and equity (AN-7, AN-8, C3)

With stock on 2025-12-31, closing every district's gap at the planning quantile would cost 0.7 M BDT; at a budget of 60% of that (0.4 M) the optimised plan with $\alpha = 0.10$ leaves weighted unmet need of 2,174 vs 8,129 patient-equivalents for the population-proportional plan, covers 58.0% vs 26.9% of the gap, and leaves 13 vs 29 districts below half coverage; unmet rapid-response team demand is 0 vs 0 teams. Table VII gives the efficiency–equity curve over six budgets. Rule: take the smallest $\alpha$ that leaves no district with a gap and nothing shipped, provided it costs at most 10% extra weighted unmet demand against the purely efficient plan; at 60% budget this gives $\alpha = 0.10$ (loss 7.1%, districts left with nothing 3→0). **Interpretation.** In this low-season snapshot (end of December) the shortfall is almost entirely ORS and IV saline; per taka, ORS protects about four times more patient-equivalents than IV saline, so the plan first closes ORS gaps everywhere, then IV saline where alerts or growth raise the weight. Population-proportional allocation spends the same money but ships saline and sachets to districts that already hold enough, which is why it meets less than half as much of the gap. Every extra step of $\alpha$ costs efficiency with diminishing equity gains (Table VII); a small floor ($\alpha = 0.10$) is enough to guarantee every district a share, and larger values mostly move supplies from districts with a gap to districts without one.

### D. Scalability and Spark optimisations (ENG-2, ENG-3)

Table VIII reports the full `make all` (generator, load, analytics, export) at three scale factors on 4 vCPUs and 16 GB RAM (Spark local mode, 6 GB heap, Java 17): SF = 1 finishes in 22.5 minutes, well inside the 45-minute limit. Time grows roughly linearly with rows in the load stage, while analytics is almost constant because it works on weekly/daily aggregates whose size depends on districts and weeks, not on encounters. The Spark UI shows the silver stage as the bottleneck: it evaluates 16 rules, hashes 28M IDs, shuffles once by `encounter_id` for resubmission handling and writes 48 partitions. Caching the evaluated batch (reused by three

TABLE V
DETECTION OVER THE THRESHOLD GRID (* = OPERATING POINT)

| Method | Thr. | Sens. | Delay | FA/DY | FA naive | NRS |
|---|---|---|---|---|---|---|
| EARS C1 | 1.5 | 100% | 2 | 80.41 | 80.39 | 100% |
| EARS C1 | 2 | 97% | 5 | 53.08 | 53.06 | 100% |
| EARS C1 | 2.5 | 94% | 6 | 33.59 | 33.59 | 100% |
| EARS C1 | 3 | 91% | 7 | 21.46 | 21.36 | 100% |
| EARS C1 | 3.5 | 81% | 8 | 13.92 | 13.72 | 100% |
| EARS C1 | 4 | 78% | 11 | 9.48 | 9.29 | 100% |
| EARS C1 | 5 | 41% | 10 | 5.06 | 4.91 | 95% |
| EARS C1 | 6 | 12% | 10 | 2.88 | 2.80 | 88% |
| EARS C1 | 8 | 6% | 4 | 0.95 | 0.95 | 62% |
| EARS C2 | 1.5 | 100% | 4 | 71.01 | 70.79 | 100% |
| EARS C2 | 2 | 100% | 5 | 49.03 | 48.96 | 100% |
| EARS C2 | 2.5 | 97% | 6 | 33.58 | 33.69 | 100% |
| EARS C2 | 3 | 91% | 7 | 23.21 | 23.20 | 100% |
| EARS C2 | 3.5 | 88% | 7 | 16.35 | 16.35 | 100% |
| EARS C2 | 4 | 88% | 8 | 11.70 | 11.60 | 100% |
| EARS C2 | 5 | 66% | 8 | 6.86 | 6.77 | 97% |
| EARS C2 | 6 | 53% | 7 | 4.54 | 4.53 | 97% |
| EARS C2 | 8 | 34% | 14 | 2.48 | 2.51 | 84% |
| EARS C3 | 1.5 | 100% | 4 | 40.68 | 41.00 | 100% |
| EARS C3 | 2 | 100% | 6 | 32.13 | 32.21 | 100% |
| EARS C3 | 2.5 | 100% | 7 | 25.45 | 25.37 | 100% |
| EARS C3 | 3 | 97% | 7 | 20.38 | 20.47 | 100% |
| EARS C3 | 3.5 | 94% | 8 | 16.48 | 16.41 | 100% |
| EARS C3 | 4 | 94% | 8 | 13.33 | 13.17 | 100% |
| EARS C3 | 5 | 88% | 8 | 9.05 | 8.99 | 100% |
| EARS C3 | 6 | 69% | 8 | 6.15 | 6.14 | 100% |
| EARS C3 | 8 | 59% | 10 | 3.19 | 3.26 | 92% |
| CUSUM | 2 | 97% | 0 | 20.88 | 20.62 | 100% |
| CUSUM | 3 | 97% | 3 | 15.11 | 15.15 | 100% |
| CUSUM | 4 | 94% | 4 | 12.15 | 12.09 | 100% |
| CUSUM | 5 | 94% | 5 | 10.08 | 10.00 | 100% |
| CUSUM | 6 | 94% | 8 | 8.17 | 8.24 | 100% |
| CUSUM | 8 | 94% | 9 | 5.92 | 5.90 | 100% |
| CUSUM | 10 | 94% | 11 | 4.41 | 4.46 | 100% |
| CUSUM | 12 | 94% | 12 | 3.41 | 3.44 | 100% |
| CUSUM | 14 | 91% | 13 | 2.74 | 2.79 | 100% |
| CUSUM | 17* | 91% | 15 | 2.02 | 2.09 | 100% |
| CUSUM | 20 | 88% | 16 | 1.53 | 1.58 | 100% |
| CUSUM | 25 | 88% | 16 | 1.03 | 1.05 | 100% |

Delay = median days to detection; FA/DY = false alert events per district-year (4 diseases); NRS = share of districts detecting the national surge.

TABLE VI
BACKTEST (LAST 52 WEEKS): MAE, 80% COVERAGE, WIS RELATIVE TO SEASONAL NAIVE

| Disease | h | MAE: naive / GLM / HGB | | | $\text{Cov}_{\text{naive}}$ | $\text{Cov}_{\text{GLM}}$ | $\text{rWIS}_{\text{GLM}}$ | $\text{rWIS}_{\text{HGB}}$ |
|---|---|---|---|---|---|---|---|---|
| dengue | 1 | 48.4 | 32.1 | 25.9 | 82% | 80% | 0.43 | 0.44 |
| dengue | 4 | 51.5 | 38.4 | 33.3 | 80% | 77% | 0.44 | 0.42 |
| diarrhoea | 1 | 55.5 | 40.8 | 38.5 | 79% | 81% | 0.71 | 0.67 |
| diarrhoea | 4 | 57.4 | 44.2 | 42.0 | 78% | 83% | 0.76 | 0.73 |
| ili | 1 | 141.8 | 51.8 | 38.8 | 77% | 80% | 0.24 | 0.19 |
| ili | 4 | 148.3 | 162.8 | 150.9 | 75% | 79% | 0.75 | 0.79 |
| malaria | 1 | 3.0 | 2.3 | 2.7 | 87% | 93% | 0.64 | 0.80 |
| malaria | 4 | 3.1 | 2.6 | 3.0 | 86% | 92% | 0.71 | 0.84 |
| all | 1 | 62.2 | 31.7 | 26.4 | 81% | 83% | 0.36 | 0.33 |
| all | 4 | 65.1 | 62.0 | 57.3 | 80% | 83% | 0.68 | 0.70 |

TABLE VII
EFFICIENCY–EQUITY CURVE (WEIGHTED UNMET NEED IN PATIENT-EQUIVALENTS; COVERAGE VALUED AT COST)

| Budget | Plan | W. unmet | Gap cov. % | Min cov. | <50% |
|---|---|---|---|---|---|
| 0.2 | $\alpha = 0$ | 8,348 | 20.0 | 0% | 32 |
| 0.2 | $\alpha = 0.1$ | 8,523 | 19.4 | 0% | 33 |
| 0.2 | $\alpha = 0.25$ | 8,795 | 18.4 | 0% | 33 |
| 0.2 | $\alpha = 0.5$ | 9,284 | 16.8 | 0% | 32 |
| 0.2 | $\alpha = 0.75$ | 9,833 | 15.0 | 0% | 33 |
| 0.2 | $\alpha = 0.9$ | 10,178 | 13.9 | 0% | 34 |
| 0.2 | pop. baseline | 11,803 | 9.5 | 3% | 41 |
| 0.4 | $\alpha = 0$ | 4,480 | 40.0 | 0% | 16 |
| 0.4 | $\alpha = 0.1$ | 4,656 | 38.8 | 0% | 16 |
| 0.4 | $\alpha = 0.25$ | 4,934 | 36.8 | 0% | 18 |
| 0.4 | $\alpha = 0.5$ | 5,446 | 33.2 | 0% | 19 |
| 0.4 | $\alpha = 0.75$ | 6,110 | 29.0 | 0% | 19 |
| 0.4 | $\alpha = 0.9$ | 7,012 | 26.4 | 0% | 24 |
| 0.4 | pop. baseline | 9,824 | 18.6 | 5% | 32 |
| 0.6 | $\alpha = 0$ | 2,030 | 60.0 | 0% | 13 |
| 0.6 | $\alpha = 0.1$ | 2,174 | 58.0 | 0% | 13 |
| 0.6 | $\alpha = 0.25$ | 2,388 | 55.0 | 0% | 14 |
| 0.6 | $\alpha = 0.5$ | 3,198 | 49.0 | 0% | 18 |
| 0.6 | $\alpha = 0.75$ | 4,199 | 42.0 | 0% | 17 |
| 0.6 | $\alpha = 0.9$ | 4,913 | 36.9 | 0% | 17 |
| 0.6 | pop. baseline | 8,129 | 26.9 | 8% | 29 |
| 0.8 | $\alpha = 0$ | 709 | 80.0 | 50% | 0 |
| 0.8 | $\alpha = 0.1$ | 864 | 76.5 | 50% | 0 |
| 0.8 | $\alpha = 0.25$ | 1,241 | 71.1 | 8% | 5 |
| 0.8 | $\alpha = 0.5$ | 1,849 | 62.6 | 3% | 11 |
| 0.8 | $\alpha = 0.75$ | 2,569 | 53.5 | 0% | 16 |
| 0.8 | $\alpha = 0.9$ | 3,612 | 46.1 | 0% | 17 |
| 0.8 | pop. baseline | 6,627 | 34.7 | 11% | 24 |
| 1 | $\alpha = 0$ | 0 | 100.0 | 100% | 0 |
| 1 | $\alpha = 0.1$ | 180 | 94.9 | 50% | 0 |
| 1 | $\alpha = 0.25$ | 450 | 87.3 | 61% | 0 |
| 1 | $\alpha = 0.5$ | 976 | 74.9 | 25% | 2 |
| 1 | $\alpha = 0.75$ | 1,873 | 62.2 | 7% | 11 |
| 1 | $\alpha = 0.9$ | 2,469 | 54.2 | 0% | 15 |
| 1 | pop. baseline | 5,386 | 41.6 | 14% | 13 |
| 1.2 | $\alpha = 0$ | 0 | 100.0 | 100% | 0 |
| 1.2 | $\alpha = 0.1$ | 0 | 100.0 | 100% | 0 |
| 1.2 | $\alpha = 0.25$ | 0 | 100.0 | 100% | 0 |
| 1.2 | $\alpha = 0.5$ | 454 | 87.2 | 61% | 0 |
| 1.2 | $\alpha = 0.75$ | 1,277 | 70.7 | 28% | 4 |
| 1.2 | $\alpha = 0.9$ | 1,999 | 60.5 | 8% | 12 |
| 1.2 | pop. baseline | 4,402 | 48.1 | 16% | 11 |

TABLE VIII
SCALABILITY: SECONDS PER STAGE, TOTAL MINUTES, DATA SIZE (GB; THE SF = 1 LAKE ALSO HOLDS THE DELTA VERSIONS OF THREE LATER INCREMENTAL LOADS)

| SF | Rows | Gen. | Bronze | Silver | Gold | Analyt. | Export | Total | Landing | Lake |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.25 | 7.1M | 21 | 66 | 163 | 48 | 374 | 50 | 12.4 | 0.95 | 0.50 |
| 0.5 | 14.1M | 34 | 86 | 293 | 54 | 407 | 51 | 15.8 | 1.68 | 0.91 |
| 1 | 28.2M | 60 | 122 | 517 | 79 | 485 | 54 | 22.5 | 3.15 | 2.69 |

actions) did not shorten it measurably on this machine (510 s vs 549 s), because the cached data spill to disk; the next step would be to fold the statistics into the write job.

Two optimisations with before/after evidence (median of three runs on the SF = 1 lake, Spark UI screenshots in `docs/experiments/screenshots`): (i) **broadcast join of the facility registry** – with broadcasting disabled Spark plans a SortMergeJoin that shuffles all 28.2M encounters (646 MB of shuffle data) and sorts both sides; broadcasting the 786-row registry gives a BroadcastHashJoin with no shuffle of the large side: 7.04 s → 4.23 s (1.7×); (ii) **columnar storage** – the same per-facility, per-diagnosis count over all encounters reads 2.94 GB of CSV in the landing zone but only two of 17 columns from 0.33 GB of zstd Parquet in bronze: 13.40 s → 1.95 s (6.9×). (iii) **Partition pruning** on silver (epi_year×division) cuts the files scanned by a division-year query from 14 to 1, but the wall-time gain is small at this size (0.50 s → 0.46 s) because the table is already in the operating-system cache; it grows with data volume. Adaptive query execution (skew-join handling, coalescing of the 32 shuffle partitions) is enabled throughout; we did not run a separate skew experiment.

TABLE IX
DENGUE PREPAREDNESS CALENDAR (NEXT 12 MONTHS)

| Division | Campaign | Pre-position | Onset | Peak | End |
|---|---|---|---|---|---|
| Barishal | 2026-03-09 | 2026-04-06 | 2026-05-04 | 2026-08-17 | 2026-11-16 |
| Chattogram | 2026-03-23 | 2026-04-20 | 2026-05-18 | 2026-08-17 | 2026-11-16 |
| Dhaka | 2026-03-16 | 2026-04-13 | 2026-05-11 | 2026-08-10 | 2026-11-09 |
| Khulna | 2026-03-09 | 2026-04-06 | 2026-05-04 | 2026-09-07 | 2026-11-16 |
| Mymensingh | 2026-03-23 | 2026-04-20 | 2026-05-18 | 2026-09-14 | 2026-11-09 |
| Rajshahi | 2026-03-16 | 2026-04-13 | 2026-05-11 | 2026-08-31 | 2026-11-09 |
| Rangpur | 2026-03-16 | 2026-04-13 | 2026-05-11 | 2026-08-31 | 2026-11-09 |
| Sylhet | 2026-03-23 | 2026-04-20 | 2026-05-18 | 2026-09-14 | 2026-11-02 |

### *E. Incremental loads, streaming and calendar*

**Incremental and idempotent (`ING-4`).** Loading the first future daily drop into the SF = 1 lake added 13,094 encounters (including late records whose onsets fall in earlier epi-weeks) in 135 s, and the whole drop DAG including analytics and dashboard export took 11.5 minutes. Re-running the same day returned "nothing new" in 18 s; a forced re-ingest (124 s) left silver at 27,267,759 rows, all with distinct `encounter_id`, and every gold count unchanged. **Hidden-drop rehearsal.** Pointing `make drop` at the folder with all 14 generated days (same schema, new outbreaks, quality defects) skipped the day already loaded and ingested the other 13 days (174,804 rows, 2,852 quarantined) in 11.4 minutes end to end, regenerating alerts, forecasts, allocation and dashboard data without any code change. **Streaming (`ING-6`).** With the Structured Streaming job running (30-second trigger), a new encounter file dropped in the inbox refreshed `alerts.json` after a median of 87 s (maximum 129 s), inside the 5-minute requirement. **Time travel (`ING-7`).** Every load creates a new Delta version of silver and gold, so "what did we know on day $t$" is a `versionAsOf`/`timestampAsOf` read; the backtest uses the reporting triangle, which gives the same answer at weekly resolution without storing a version per day. Table IX shows the dengue calendar for the next 12 months.

## VII. DASHBOARD DESIGN AND USABILITY

The dashboard (`VIS-1`) is plain HTML with D3.js v7 and topojson-client, vendored so it runs offline, and is started with

TABLE X
SEIR WHAT-IF SCENARIOS FOR THE RESPIRATORY SURGE

| Scenario | Reported | Peak wk | Peak | Averted |
|---|---|---|---|---|
| no intervention (fitted) | 354,343 | 8 | 69,551 | 0% |
| 20% contact reduction at detection | 297,155 | 10 | 39,758 | 16% |
| 20% contact reduction 2 weeks earlier | 296,791 | 10 | 39,659 | 16% |
| 20% contact reduction 2 weeks later | 298,973 | 9 | 41,148 | 16% |
| 40% contact reduction at detection | 160,018 | 14 | 11,230 | 55% |
| 40% contact reduction 2 weeks earlier | 142,842 | 17 | 10,615 | 60% |
| 40% contact reduction 2 weeks later | 184,967 | 9 | 14,995 | 48% |

one command (`make serve`). Fig. 3 shows the national view.

**Coordinated views (`VIS-2`, `VIS-3`).** (1) District choropleth of mean weekly incidence per 100,000 over the selected weeks with alert markers; (2) focus+context time series with the reported counts, the nowcast for weeks still being reported, forecast median with 50/80/90% bands and the alert threshold (expected 7-day count + 2 SD); (3) seasonal heatmap, epi-week×year; (4) allocation view comparing forecast demand (10–90% interval and planning quantile), stock, recommended and baseline shipments; (5) ranked early-warning table; plus the what-if panel, the preparedness calendar, SEIR scenarios, the per-load data-quality report and an evaluation panel showing the detection trade-off and forecast skill. One shared state (disease, area, week range, resource, threshold, budget, $\alpha$) is broadcast to every view with `d3.dispatch`: clicking a district on the map, a row in the table or a bar in the allocation view selects it everywhere; brushing the context strip or clicking a heatmap cell sets the period of the map and heatmap. Every view has tooltips, legends and axis labels.

**Scale-aware design (`VIS-4`).** The browser loads only the role's bundle of aggregated JSON: the national bundle is about 1.8 MB plus a 55 kB TopoJSON (limit 10 MB, checked by a test). Counts are at district×epi-week (the level at which district officers act and stocks are managed); forecasts, demand and allocations are per district for four weeks; detection is published as daily *statistics* for the last 180 days so the threshold can be changed in the browser without shipping counts.

**What-if panel (`VIS-5`).** Sliders for budget (20–120% of the gap), alert threshold and equity floor $\alpha$ look up a precomputed grid of optimiser runs (every combination solved offline, because the alert threshold changes the weights and team demand) and recompute alert status from the published statistics; the efficiency–equity curve highlights the current $\alpha$.

**Accessibility and honesty (`VIS-6`).** Disease colours use a validated colour-blind-safe order (blue, orange, aqua, yellow; adjacent CVD $\Delta E \geq 9$ in both light and dark mode, checked with a palette validator), magnitudes use a one-hue blue ramp, alert states use reserved status colours always paired with an icon and a label; suppressed cells are hatched and shown as "<5"; every forecast carries its bands; every panel shows its data-freshness date and a banner states that the data are synthetic and which public inputs used a fallback. **Mobile and playback (`VIS-7`).** The grid collapses to one column below 1,100 px and a "Play spread" button animates weekly incidence over the last year. **Roles (`PRV-3`).** A district officer's token returns a bundle with their own district in detail and all other areas as division aggregates.

**Usability check.** [The protocol in `docs/usability/usability_protocol.md` (six tasks, think-aloud, System Usability Scale) must be run with at least two non-CS users; insert their results and the changes made here.]

## VIII. Ethical, Societal and Sustainability Considerations

**Consequences of errors (A4).** A missed outbreak costs lives; a false alarm costs money and, repeated, the trust that makes officials act on the next alert. We therefore report both error types, choose the operating point openly, route reporting-artefact alerts to "verify reporting" instead of "alert", and recommend that every alert is confirmed with the district before teams move. **Privacy.** The design follows minimisation and purpose limitation: identifiers stop at bronze, salted hashes cannot be reversed without a key that never enters the repository, exact ages are dropped, and the public-facing layer contains only suppressed aggregates; stigma is a real risk for small communities (e.g. malaria in hill districts), which is why small counts are hidden even at district level and why we recommend differential privacy for public releases. **Equity.** Allocating purely on expected unmet need concentrates supplies in Dhaka and the hill districts; the equity floor makes the trade-off explicit and gives policymakers a number to defend ("every district gets at least $\alpha$ of its population share"). Weights for alerts and growth are stated and adjustable; they must be agreed with the Ministry rather than set by engineers. **Misuse.** Alerts could be used to blame districts or facilities; the dashboard shows reporting context (back-fills, new facilities) to avoid penalising good reporting, and access is role-based and audited. **Automation bias.** Forecasts and plans are advisory: uncertainty bands, the baseline plan and the what-if panel invite human judgement; the model is interpretable by design (C5). **Sustainability.** The whole platform runs on one laptop-class machine in about 22.5 minutes per full rebuild and a few minutes per daily increment, uses open-source software only, stores data compressed (zstd Parquet) and recomputes only affected epi-weeks, keeping energy and cost low enough for district infrastructure; the same code scales out to a cluster or object store without changes. **Honesty about data.** Synthetic data can make methods look better than they will be on real reports (cf. the Google Flu lesson [6]); all results must be re-validated on Ministry data.

## IX. Limitations and Future Work

(1) *Synthetic data.* Outbreak shapes, delays and artefacts follow our own generator, so detection and forecast accuracy are optimistic; validation on historical DGHS surveillance data (e.g. the 2019 and 2023 dengue seasons) is the first next step. In this build weather and population were synthetic fallbacks because the build machine could not reach Open-Meteo and HDX; the pipeline fetches the real data when run with internet access. (2) *Detection.* Weather-driven genuine rises (post-flood diarrhoea, early dengue seasons) count as false alerts against a ground truth that lists only injected outbreaks; a covariate-adjusted baseline (e.g. the forecast GLM's residuals) and Farrington-flexible [2] are natural extensions; slow outbreaks are partly absorbed by the adaptive level. (3) *Forecasting.* Direct models are refitted every four weeks, features at training time use complete data while prediction uses nowcasts, and intervals assume independent weeks. (4) *Optimisation* is single-period with linear costs; multi-period stock dynamics, lead times, cold-chain constraints and uncertainty (stochastic or robust programming) are future work. (5) *Privacy.* Complementary suppression protects the published hierarchy but not arbitrary external data; formal differential privacy is recommended for public releases. (6) *Engineering.* Local-mode Spark on one node; HDFS/Kubernetes deployment, upazila-level maps with stronger suppression, SMS alerts to district officers and integration with DHIS2 APIs are future work.

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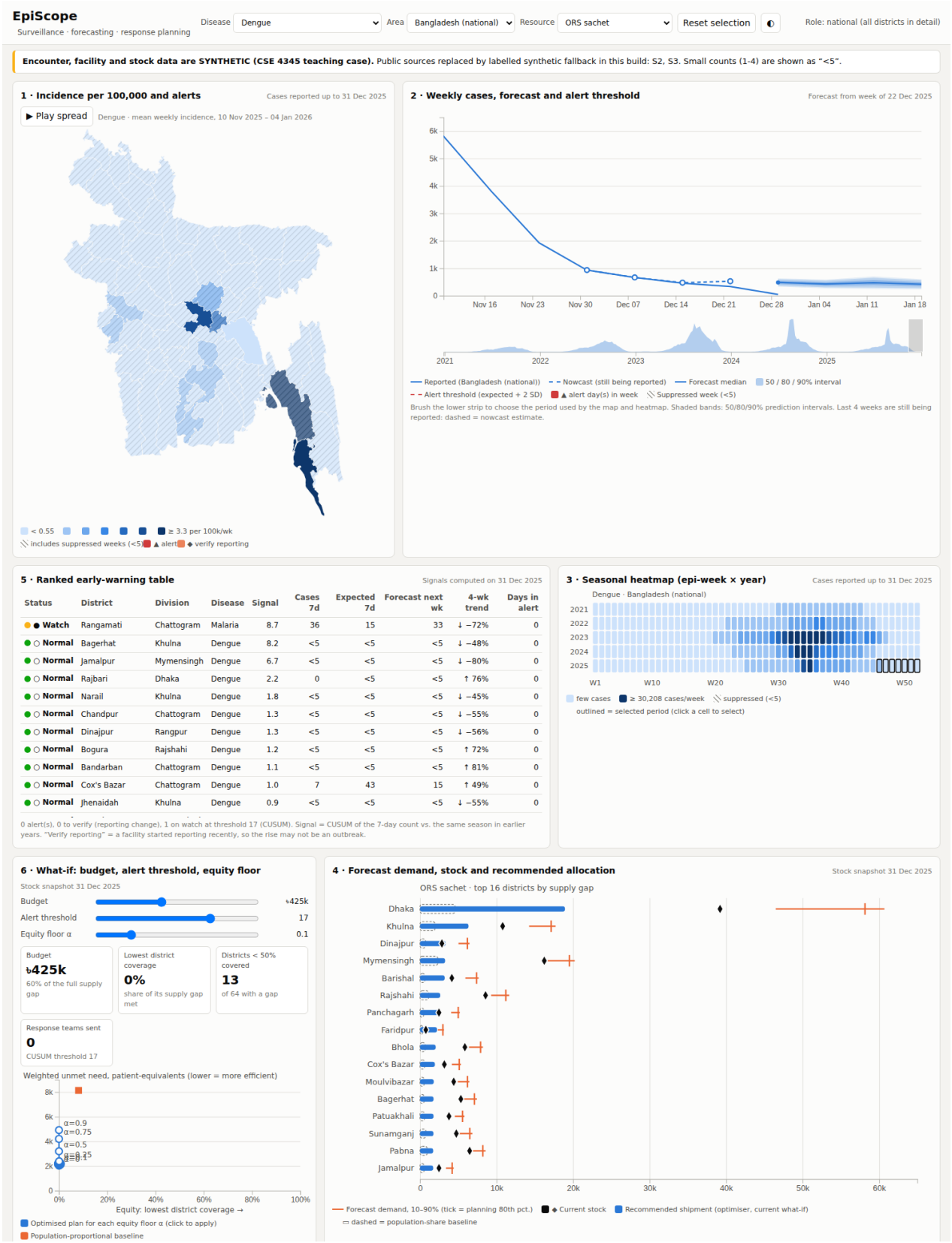



Fig. 3. EpiScope dashboard, national role (SF = 1 build). Views are linked: selecting a district on the map updates the time series, heatmap, table and allocation view.

APPENDIX A
POLICY BRIEF FOR THE DIRECTOR, DISEASE CONTROL & PREPAREDNESS UNIT

# Seeing outbreaks sooner and sending supplies where they save most lives

EpiScope prototype – summary for decision-makers

**Important:** this prototype was built and tested on realistic *simulated* patient records for all 64 districts (2021–2025), because real patient data cannot be used for a student project. The findings below show what the system can do; the numbers must be re-checked on the Ministry's own data before any operational decision.

*1. The current situation*

Information about dengue, diarrhoea and cholera, influenza-like illness and malaria reaches the Ministry late, in many formats, and is mostly analysed after the fact. As a result, saline, test kits and teams often reach a district after its hospitals are already full. EpiScope brings the routine reports from hospitals and health complexes, the weather record, population figures and district stock levels together every day and turns them into four things officials can act on: **early warnings**, **forecasts for the next four weeks**, **a recommended distribution of supplies and response teams**, and **a calendar for the year ahead**.

*2. Top three recommendations*

1) **Adopt a weekly "warning review" based on the ranked warning list.** In our test the warning system recognised 91% of the simulated local outbreaks, typically about 15 days after they began, while producing roughly 2.02 unnecessary warnings per district per year across the four diseases. It also ignored a hospital that suddenly sent three weeks of late reports at once – a situation that fools simpler systems. *Expected impact:* response teams can be sent days to weeks earlier than with monthly spreadsheet reviews, with a manageable number of false alarms. Each warning should be confirmed by a phone call to the district before a team is sent.
2) **Distribute supplies with the recommended plan, guaranteeing every district a small minimum share** (at least one tenth of its share by population of every shipment). With a budget of 60% of what would be needed to fill every district's shortfall, the recommended plan fills 58.0% of the districts' combined shortfall, against 26.9% if supplies are shared by population alone, and guarantees every district at least a share of each shipment. This guarantee costs a little efficiency (about 7.1% more unmet need than a plan that ignores fairness) but means no district is left with nothing. *Expected impact:* more saline and test kits where patients will actually be, without abandoning low-risk districts.
3) **Follow the preparedness calendar: start dengue prevention in March and stock up by April.** Every year dengue cases in all divisions begin to rise in May, peak in August and fall in November; diarrhoea peaks in April–May. The calendar tells each division when to begin larval control (about eight weeks before the rise), when supplies must be in place (four weeks before) and when to add hospital staff (two weeks before the peak). *Expected impact:* the heaviest weeks of the season are met with stock already in place.

*3. How to read the dashboard*

- **Map:** darker blue = more cases per 100,000 people in the chosen weeks. A red triangle marks a district with an active warning; an orange diamond means "check the reporting first" (for example, a new hospital has just joined the system). Striped areas include small numbers that are hidden for privacy.
- **Weekly chart:** the solid line is what has been reported; the dashed end is our estimate for the weeks still being reported; the shaded fan is the forecast – the darkest band is where the true number is most likely to fall, the lightest covers 9 out of 10 outcomes. The red dotted line is the warning threshold.
- **Warning list:** districts at the top need attention first. "Days in alert" shows how long the warning has lasted.
- **Supplies chart and sliders:** compare expected need (orange), current stock (black diamond) and the recommended shipment (blue). The sliders let you ask "what if the budget were smaller?" or "what if we guarantee every district a larger minimum share?" and see the effect immediately.
- Every panel shows the date of its latest data. A district officer sees their own district in full and the rest of the country only in summary.

*4. Confidence and limitations*

- Forecasts one week ahead were much closer to what happened than simply assuming "same as this week last year": their error score was only 0.36 times that of the simple method, and the stated ranges contained the real number about as often as promised (8 times in 10). Forecasts become less certain further ahead, and the fans in the chart show this honestly. Four-week forecasts should guide preparation, not exact orders.
- Slow-building outbreaks are harder to spot than sudden ones; unusual weather (floods, an early monsoon) also causes genuine rises that the system will flag and that officials must interpret.

- The results depend on hospitals reporting regularly. The system detects facilities that stop reporting, but it cannot see patients who never reach a facility.
- Supply needs are converted from cases using standard assumptions (for example, six bags of saline per admitted dengue patient); these should be confirmed with clinicians and stores staff.
- Some public inputs (weather, population) must be refreshed from the official sources when the system is installed at the Ministry.

*5. Data-protection assurance*

- Names and national ID numbers never leave the secure first storage area: they are replaced by an irreversible code made with a secret key that is kept outside the software.
- Exact ages are reduced to age groups. The dashboard and every export show only totals per district and week; any count between 1 and 4 is shown as “fewer than 5”, and a second number is hidden when totals would otherwise reveal it.
- Access is by personal key, each user sees only what their role requires, and every access is logged.
- Any record-level file shared with researchers is checked so that each combination of place, time, age group and sex covers at least five people; for public releases we recommend adding a small amount of statistical noise (differential privacy).

## Appendix B
## Contribution Matrix

To be completed by the group: enter L (lead), S (support) or – (no involvement) for each member and area, agree the overall percentage shares (total 100), copy the commit counts from `git shortlog -sn` and sign. The individual moderation factor is $\pi_i = \text{clip}(3\sigma_i/100, 0.8, 1.1)$. The matrix must be consistent with the Git history and with `AI_USAGE.md`.

| **Work area** | **[Member 1]** | **[Member 2]** | **[Member 3]** |
|---|---|---|---|
| Problem analysis, proposal, conflict analysis | | | |
| Synthetic generator and public-data acquisition | | | |
| Ingestion, lakehouse layers, incremental loads | | | |
| Data quality, lineage and privacy controls | | | |
| Outbreak detection and evaluation | | | |
| Forecasting, backtesting and demand conversion | | | |
| Optimisation model and preparedness calendar | | | |
| Serving layer and D3.js dashboard | | | |
| Scalability experiment and Spark optimisations | | | |
| Report and policy brief | | | |
| Video production | | | |
| **Agreed overall share $\sigma_i$ (%, total 100)** | | | |
| Commits / lines changed (from `git shortlog -sn`) | | | |
| Signature | | | |

## Appendix C
## Individual Reflections

Each member writes their own half-page reflection in their own words (it is graded individually under R9 and must not be generated). Suggested prompts: what you led and what you contributed outside your role; the hardest technical problem you solved and how you know the solution is right; one design decision you would change and why; what you learned about teamwork and about communicating with non-technical users; how you used AI tools and what you had to verify or correct yourself.

*[Member 1 name]* – *Data Engineer*

[Reflection (about 250 words).]

*[Member 2 name]* – *Analytics Engineer*

[Reflection (about 250 words).]

*[Member 3 name]* – *Visualisation Engineer*

[Reflection (about 250 words).]